



How has process assessment been automated by organizations?

A systematic literature mapping

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ABSTRACT

Organizational process improvement initiatives typically apply some kind of process assessment. Even partial automation of these process assessments can be beneficial. In this sense, the aim of this study is to investigate the state of the art in automated process assessment. A systematic literature mapping is performed in order to answer the research question: What are the approaches to automated process assessment? Thus, the state of the art is analyzed following the steps of protocol definition, search execution, study selection and data extraction and synthesis. We found 16 studies that directly or indirectly use some kind of automated process assessment support. The approaches reported in these studies cover most of the steps expected in typical process assessments and used nine different software tools to support these steps. The main observed results of process assessment were reduction of effort, time and complexity and automated assessment feedback. There are two major contributions of this research: a comprehensive and structured overview of the state of the art in automated process assessment, contributing to the knowledge organization of this field; and an initial guide to organizations that want to apply efforts to automate their process assessments in choosing the tools, techniques and approaches, and knowing the expected outcomes of this kind of initiative.

CCS CONCEPTS

• **Information systems** → Information systems applications; Process control systems.

KEYWORDS

Process Assessment, Literature Mapping, Automated Assessment

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1 INTRODUCTION

Information systems are intricately linked to process management in an organization, as business processes typically, at least in parts, demand process automation or support through IT infrastructure during its evolution [1]. There are many positive effects of a process-oriented management in an organization, such as financial performance, increased process speed and improvement product quality, as well as productivity and customer satisfaction [2]. However, there are also many difficulties in evolving process management, such as organizational culture, lack of support among senior management, the absence of clear roles and responsibilities in implementing the methodology, and insufficient budget and available resources [3].

These difficulties can be overcome when organizations achieve a high level of process maturity [4]. Process maturity has been assessed using process maturity models as a reference [5]. There are many initiatives proposing Process Maturity Models (PMMs) [6], with varied focus and depth, such as the generic Business Process Maturity Model - BPMM [4], defined by the Object Management Group (OMG).

Process assessments can be performed using a Process Assessment Model (PAM) [7]. PAM is composed by (i) a Measurement Framework, which defines capability and/or maturity levels, process attributes and rating criteria, and (ii) a Process Reference Model (PRM), which defines the processes and expected results. Process Maturity Models can often be classified as PRMs, as well as other more generic models, not necessarily formally defined process maturity models, can also be considered as PRMs, as long as they indicate processes and expected results minimally.

However, despite its recognized benefits, process assessments can be complex, expensive and time and resource consuming [8] [9]. Performing process assessments without the support of a software tool, tends to generate intrusive interruptions in the process execution, limiting the frequency of assessments and limiting the standardization of the assessment process [6] [10]. Considering

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that many expected results of PRMs can possibly be collected and measured without the need for human intervention, there is an opportunity towards the automation of process assessments.

This opportunity becomes the motivation to understand what are the approaches to automated process assessment and what would be its possible benefits. In order to answer these questions, this paper presents a systematic literature mapping of primary studies that report approaches, experiences and tools related to process assessment automation and its observed results.

The contribution of this paper is twofold: first, to the best of our knowledge, this research presents a comprehensive and structured view of the state of the art on the automated process assessment, raising indications of its possible benefits based on the search and analysis of the primary studies; second, the systematic mapping results can guide organizations that wish to apply efforts in automating the assessment of their processes, in the choice of tools, techniques, approaches and expected results of this type of initiative.

The remainder of this paper is organized as follows: section 2 presents the theoretical background on automated process assessment and section 3 presents the methodological approach used. In section 4 the search protocol is defined, followed by the execution of the search. In section 5 the data are synthesized, and the results discussed, followed by section 6 where our conclusions are summarized.

2 BACKGROUND

Process assessment can be defined as a technique to understand the current capability/maturity level of business practices, usually performed to draw and track a roadmap for process improvement [6]. Process assessments typically use some process model as a reference in order to be able to compare the actual process of an organization with that reference model. Several Process Maturity Models (PMM) have been proposed as reference for process assessment, some of them being general in scope, theoretically applicable to any business area, and others to specific business areas [11] [12].

In general, PMMs typically contain two dimensions: a Process Reference Model and a Measurement Framework [13]. Process Reference Model (PRM) defines a set of processes, their objectives and expected process outcomes [13] and the Measurement Framework contains the process assessment method, the assessed process attributes, capability and/or maturity levels, as well as the process measurement scales. Both dimensions form the Process Assessment Model (PAM) [13].

A Measurement Framework defines criteria that indicate the ability of a process or an organization to achieve its business objectives, achieving the process attributes associated with each level [7]. Assessing an organizational process therefore comprises using a PAM to assess the alignment of each process to a given PRM. This assessment is typically performed in a manual manner, analyzing the evidence left by the execution of the process in relation to the criteria established in the chosen PRM.

When organizations decide to perform a process assessment, the most prevalent approaches are usually those based on Self-Assessment or Team-led Assessment [14]. While Team-led Assessment approaches are performed by internal or external specialized



Figure 1: Research methodology, based on [15] [16]

teams using their selected assessment methods and models, in the Self-Assessment approaches the assessment is usually performed without deep experience or training in process assessment, resulting in different degrees of formality and assessment results between the two approaches. However, an approach to process assessments that has been gaining ground, especially among software-development organizations, is the Automated Process Assessment that complement or often replace Self-Assessments or Team-led Assessment approaches [10].

Process assessments can be fully automated or partially automated. In a fully automated assessment, without human intervention, the process indicators are integrated into the execution of the business process and the assessment uses criteria that help in the interpretation of the measurement data. These criteria are typically included in a software tool that supports the execution of the process. In the partially automated assessment, audits are still performed, but they are supported by pre-obtained process indicators, that are used as evidence for the assessment.

Automated assessments, even if partially, tend to bring several benefits, such as: resources saving, reducing intrusive interruptions in the execution of the processes, increasing the frequency of assessments and standardization of the assessment method ensuring the reproducibility of assessments [10]. Automated process assessments can also deliver immediate results, without the need to cyclically interrupt the process execution to perform formal audits.

Based on these assumptions, automated assessments can possibly provide detailed, real-time information to meet the needs of tracking process alignment with reference models in a timely manner, supporting and facilitating continuous process improvement.

3 METHODS

A Systematic Literature Mapping (SLM) is a literature review method used to structure a field of interest, providing an overview of a research area, and identifying the quantity and type of research and results available within it [15].

In order to provide an overview on the current state of the art on automated process assessment, we performed a Systematic Literature Mapping based on the procedures proposed by Petersen et al [15] and Kitchenham & Charters [16]. The review was performed by the authors, following the steps in Figure 1.

Following the SLM approach, a series of activities were organized into four main phases: protocol definition, where the research is planned; execution of searches, applying the search strings to the selected digital bases; study selection, where the primary studies are selected according to the defined inclusion and exclusion criteria; and data extraction and synthesis.

Each phase is constituted by a set of steps which were adopted in this research and are presented below, in a grouped form, in the following sections: (4) protocol definition, searches execution and study selection and (5) data extraction and synthesis.

Table 1: Search String

((("automation" OR "automatics" OR "tool" OR "semi-automated" OR "semi automated" OR "software" OR "automated") AND ("process evaluation" OR "process assessment" OR "process audit" OR "process appraisal" OR "process self assessment" OR "process model" OR "process standard"))))

4 PROTOCOL DEFINITION

This section presents the protocol definition, which is necessary to reduce the possibility of researcher bias, by specifying the methods that will be used to undertake a specific systematic review [16].

4.1 Research Question

From the identified research need, the general research question was defined as: "What are the approaches to automated process assessment?".

From this research question, we are interested in finding studies that indicate some kind of automated process assessment, even if partial and/or informal, so that interesting data about the process assessment automation can be collected. We refined the main research question into the following analysis questions:

- AQ1: How are the automated process assessment approaches defined?
- AQ2: What are the tools used to support automated process assessment?
- AQ3: What are the process reference models or standards used for automated process assessment?
- AQ4: How are the results of the automated process assessment?

4.2 Data source and search strategy

Searches were carried out in following digital libraries: IEEEExplore, ScienceDirect, Springer and ACM Digital Library. In order to select the primary studies, inclusion and exclusion criteria are defined, based on [15] and [16]:

Inclusion criteria: Studies that present methods, techniques or tools for automated processes assessment; Studies fully available through the selected scientific databases; Studies published after 2005; Papers that present empirical studies with results on automated process assessment.

Exclusion criteria: Studies that do not address process management even in an informal perspective; Studies that address just process-automation tools but not process assessment.

Definition of search string. Search terms were derived from the general research question and the analysis questions. In Table 1 we present the search strings containing the defined search terms, that was adapted to the appropriate syntax for each digital library.

4.3 Search execution

For the selection of the studies, three cycles were performed, applying the refinement and elimination of studies based on inclusion and exclusion criteria. Only automatic searches were performed. No manual search and selection techniques, such as snowballing, for example, were used.

The search string was adapted to the specific syntax and submitted to each selected digital library: IEEEExplore, Science Direct,

Table 2: Results by Digital Base

Digital Base	1. Iteration	2. Iteration	3. Iteration
IEEEExplore	1870	11	9
ACM Digital Library	1435	6	3
Science Direct	173	17	2
Springer	2328	3	2

ACM Digital Library and Springer [15]. The initial returned studies were then divided among the authors.

In the first iteration, titles of the returned studies were read, applying the inclusion and exclusion criteria, eliminating the irrelevant studies. In the second iteration abstracts were also read, applying again the criteria. In this second iteration, all the selected studies were validated in pairs by researchers, solving possible conflicts or individual bias. Repeated studies were also excluded.

In the third Iteration, the studies selected in the previous iteration were read integrally and confronted with the research question and also with the inclusion and exclusion criteria. Finally, from the studies considered relevant to the review, data were extracted. Table 2 shows the numbers of results per iteration. The complete list of selected studies is presented in Table 3.

5 SYNTHESIS AND DISCUSSION

This section presents the synthesis of the data collected from the selected studies, organized by analysis question. We then discuss the implications of these collected results.

AQ1: How are the automated process assessment approaches defined?

In our search we found 16 studies containing approaches that, in one way or another, contemplate some form of automated process assessment in relation to its alignment with a PRM.

Most of these selected studies do not present process assessment from such a narrow perspective, but generally show how organizations have sought to automate their process assessment. As a result, they do not present a sufficient degree of formalism to precisely define the structure of the assessment approaches applied, which would make it easier to compare the approaches used in the studies. Thus, in order to analyze the activities carried out in the assessment approaches, ISO/IEC 33020 [13] standard (formerly ISO/IEC 15504-3) was used as a reference to generalize the activities used within the selected approaches, as shown in Table 3.

Among the typically expected steps of a formal process assessment, almost all the studies indicate that they have used some form of assessment planning (~94% - 15), while most (~69% - 11) do not, even though informally, perform some form of data validation.

Table 3: List of studies classified according to ISO/IEC 33020 assessment activities.

Study	Activities				
	Planning	Data Collection	Data Validation	Process Attributes	Report Results
Calabró, Lonetti & Marchetti [17]	N	F	P	F	N
Khokhar et al [18]	F	F	F	F	F
Garcia & Pacheco [19]	F	N	N	F	F
Choi, Kim & Park [20]	F	F	F	F	F
Patel & Ramachandran [21]	F	F	P	F	F
Pino, Garcia & Piattini [22]	F	F	F	F	F
Astorga-Vargas et al [23]	F	P	P	F	N
Sabato & Filangieri [24]	F	N	N	F	N
Adali, Top & Demirors [25]	F	F	F	F	F
Homchuenchom [26]	F	N	N	F	N
Dai, Bai & Zhao [27]	F	N	N	N	N
Pino, Garcia & Piattini [28]	F	P	P	F	F
Zaouali & Ghannouchil [29]	F	P	P	P	N
Varkoi [30]	F	F	F	F	F
Suteeca & Ramingwong [31]	F	N	N	N	N
Montenegro & Arévalo [32]	F	N	N	P	F

Some of the approaches do not formally automate data collection (50% - 8) or the assessment of process attributes (25% - 4), so it can be said that they generally use a semi-automated process assessment approach. In terms of coverage of typical activities, the most complete studies [20], [22], [25] and [30] can be highlighted.

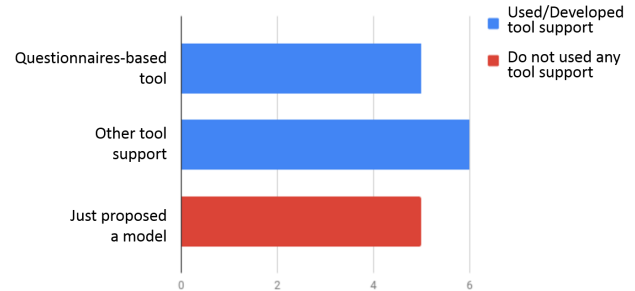
Table 3 presents selected studies/approaches and process assessment activities (ISO/IEC 33020), where “F” indicates that the activity is fully covered, “P” indicates partial coverage and “N” indicates no coverage.

Calabró, Lonetti & Marchetti [17] use an integrated framework that allows the modelling, execution and analysis of business processes based on a flexible and adaptive monitoring infrastructure, allowing the definition and assessment of user-defined KPIs (Key Performance Indexes). Similarly, Dai, Bai & Zhao [27] take a workflow-based approach to specifying and verifying web service workflow compositions by transforming a BPEL model into a TPPN template.

Adali, Top & Demirors [25] presents an online approach that has the ability to guide and automate processes for assessing modern agile organizational environments, including planning, conducting, and reporting activities in the assessment process, in order to provide a comprehensive way to assess agility without relying on a specific agile model.

Sabato and Filangieri [24], instead, bases their approach on identifying operational problems, capturing knowledge and reducing time analysis, supporting the process owner during his or her diagnostic activities. For each quality indicator, they propose a questionnaire with a set of questions, ordered by weight, oriented to identify the specific organization processes weaknesses.

Homchuenchom [26] approach uses Scrum tasks, selected and modelled with BPMN on a BPMS platform. However, the study does not address the automated process assessment model itself, but rather proposes an assessment model to generate a detailed understanding and vision of the process, facilitating assessments.

**Figure 2: Tool support**

AQ2: What are the tools used to support automated process assessment?

Most of the studies (~69% - 11) used some kind of software tool/platform in order to perform processes assessment. Others (~31% - 5), however, have proposed only a model for automated process assessment and/or improvement. Figure 2 highlights the number of studies that used or did not use tools.

One of the studies that uses a questionnaire-based approach, is the ASSO (a web application for operational process assessment and organizational improvement) [24]. Using ASSO tool, organization employees answer an electronic questionnaire, and the authors argue that the results identify the actual operational problems, reducing the analysis time. Another example is SPIALS tool (Software Process Improvement Adaptive Learning System) [26] which allows organizations to collect data on process quality and acceptance through questionnaires.

In addition to questionnaires, some studies have used or developed more complete tools to automate process assessment fully or partially. One of these tools is the SysProVal [19], which allows

Table 4: Tools used to process assessment support.

Tool	Study
Glimpse: flexible and configurable monitoring infrastructure	Calabró, Lonetti & Marchetti [17]
BPEL4WS: Business Process Execution Language for Web Services to an extended TPPN	Dai, Bai & Zhao [27]
SPIALS: Software Process Improvement Adaptive Learning System	Homchuenchom [26]
AssessAgility: web based agility assessment tool	Adali, Top & Demirors [25]
SPQA.web: a support tool for the execution of rapid process assessments	Pino, Garcia & Piattini [22]
ASSO: a Web-portal application for operational process assessment and organizational improvement	Sabato & Fllangieri [24]
GENESIS: management and implementation of all activities of a process improvement cycle	Pino, Garcia & Piattini [28]
Automated Tool Support	Patel & Ramachandran [21]
SysProVal	Garcia & Pacheco [19]

comparing organization's current practices with PRM practices, performing the assessment of the selected process and indicating a suitable process improvement plan. Other tools are web-based tools and have been developed for online assessment, such as the AssessAgility assessment tool [25]. Some studies have not developed a specific tool, but used an existing tool, such as the WofBPEL tool [27]. Tools explicitly cited in 9 studies (62%) are presented in Table 4.

Figure 1 is an example of a figure and caption spanning the half-page width (one column in a two-column format) with the styles applied. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Although not all the studies explicitly present the used tool to automate the processes assessment, most proposed a model for their case studies. In a comprehensive way, all models present the steps of planning and management, data analysis, and formulation of improvements or recommendations that could be automated. One of the models that highlights all these steps is the Agile Maturity Model (AMM) [21].

AQ3: What are the process reference models or standards used for automated process assessment?

We analyzed the process quality reference models used in process assessment. A total of 22 reference models were used in the studies as some studies used more than one model, and others did not explicitly indicate any reference model.

Among the reference models, CMMI is used by 6 studies ([18] [19] [21] [22] [24] [26]). Scrum model was used by 5 studies ([25] [26] [29] [31] [32]). ISO standards were also used by 5 studies, ISO/IEC 15504 is presented in 3 studies ([20] [22] [28]) and ISO 29110 used in two studies ([30] [31]). COBIT ([32]), AgilityMod ([25]) and COMPETISOFT ([28]) appeared in only 1 study each.

More than one third of the studies (~37% - 6) performed process assessments based on lightweight or agile models (Scrum and AgilityMod), which raises indications of a concern with process alignment, even for agile-oriented organizations. However, in (~19% - 3) of the studies, reference models were not explicitly identified, which raises indication that not all process assessments follow reference models or standards, some of them just follow, sometimes informal, homemade process reference models.

It is interesting to note that most primary studies do not use traditional and generic process quality models, such as BPMM [4], but used more specialized models for their specific business domains, mostly related to software development. This should be justified by the trend that has occurred in recent years of specializing large generic process maturity models for business domains with specific needs [6] [12] [33] [34].

AQ4: How are the results of the automated process assessment?

A large part of the studies (~56% - 9) do not explicitly indicate data on their practical application or what results have been observed from the application of automated process assessment.

Regarding the context of the use of the approaches, among studies that reported having performed practical applications, most indicated that the process assessment approach was applied in Very Small Enterprises (VSEs) (~18% - 3), being the same number of applications in medium-sized companies (~18%) and only 1 application (~6%) in a small company.

As the process assessment automation is typically applied in the context of a process improvement program within the organization, it is difficult to separate the observed results of the tool support from those resulting from the process improvement itself. In addition, most of the reported practical experiences have not been systematically evaluated, relying largely on informal observation and discussion or interviews with participants, which increases threats to the validity of their findings.

Among the studies that were applied in practice, the main results observed from the automation, even partial, raise evidence that automated process assessment: easier process diagnostics ([19] [22]), reduces assessment time and effort ([22] [25] [24]), generates automatic process improvement feedback and recommendations ([20] [22]), is suitable for small businesses ([28]) and that organizations involved had moved from a chaotic and unpredictable process to a tangible one ([23]).

On the other hand, some reports note possible weaknesses, such as that business managers tend to be skeptical about automatic process assessment, requiring more evidence of payback before becoming being fully convinced of the usefulness of the approach ([21]).

5.1 Discussion

The primary studies found in this research indicate the importance and focus on studying process assessment and, in addition, its automated assessment. However, as can be seen in Table 3, process assessment automation has not been completely covered, with “Planning” and “Assessment of process attributes” being the most common assessment activities among the 16 found studies. Yet, there are other perspectives to process assessment, that do not necessarily use those types of activities, but follow other approaches for business process measurement ([35] [36]). Those alternative approaches, however, typically covers only process performance assessments, focusing mostly on the redesign of the BPM life cycle, evaluating processes’ best practices ([37]), which is perhaps why they do not focus on automation of process assessments, and, consequently, did not appear as results in our research.

Given the difficulties in carrying out process assessments, we collected useful insights and alternatives for organizations that plan to invest efforts in automating the assessment of their processes. To assist in decision making to invest or not in the process assessment automation, we found several benefits resulting from the automation, such as obtaining easier process diagnostics ([19] [22]), reduction of assessment’s time and effort ([22] [25] [24]), and the generation of automatic process improvement feedback ([20] [22]).

In terms of the assessment process itself, instead of defining their own home-made assessment activities, organizations may choose to adopt a more comprehensive and previously tested framework, such as those proposed by Choi, Kim & Park [20], Khokhar, Mansoor, Rehman & Raufa [18] or Pino [28]. Organizations may also use our conclusions that one of the most common weaknesses (~69%) of these assessment automation processes is the lack of a validation step for the assessment data, to invest efforts to also include this step in their assessment processes.

Since most studies (~69%) use some kind of software, tool or platform to automate processes assessments, this indicates that it may be interesting for organizations to invest in software tools to support the entire assessment process. In this case, organizations can choose one of the simplest tools, as suggested in studies such as by Sabato and Philangieri [24] or Homchuenchom [26] or one of the more complete tools as suggested by Garcia [19] or Adali et al. [25]. Organizations can also adapt an existing tool, as suggested by Dai [27], or even use already proven assessment steps to implement their own tool, as proposed by Patel and Ramachandran [21].

Also, instead of using the classic large and generic process maturity models as a reference for process assessment automation, organizations could seek customizations of these models for their specific business domain, as more than 77% of primary studies followed this path. Especially for software-related companies, we found several alternatives of specific reference models that have been used for this purpose, such as CMMI (~37%) or Scrum (~31%). Organizations of other areas have also the option to first define their own customized process maturity models, following Buglione et al., [33] or Hauck et al [34], and then use them as a basis for process assessment automation.

During organizations’ efforts to automate process assessment, it is common to invest in graphically modeling their current processes

in order to gain process visibility [38]. Our results indicate that the notation languages most used for this purpose have been BPMN (~12%) and UML (~6%). Primary studies that show useful examples of this use are Calabró et al. [17] and Zaouali et al. [29].

One of the typical motivations for adopting process assessment automation is for continuous improvement. Business processes that are continuously assessed tend to be continuously improved [39], as it helps to detect deviations, as well as to discover improvement opportunities [40], as suggested by the highest maturity levels of some classic process quality models [4]. Thus, organizations can use automated process assessments as an instrument to support their process improvement. However, it was possible to notice in the primary studies that a variety of process maturity models were adopted, but no specific metrics, indicators or measurement models were identified for process improvement. In consequence, it is possible to argue that it is difficult to perform process improvement without a predefined set of metrics or performance indicators, even with the support of an automated process assessment. Therefore, this may indicate that there is a need for a broader view of process assessment automation, where process performance measurement models can be used to support process improvement, closing the link between organizational maturity and business process performance seems to be needed [41] [42]. Therefore, it can be inferred that some process-appropriated business assessment model needs to be considered in advance, so that automated process assessment solutions can be designed in a subsequent step.

5.2 Threats to Validity

Some threats to the validity of the results presented are observed, as in all systematic literature research [43]. We identified potential threats and applied mitigation strategies in order to minimize their impact on our results.

Incorrect or incomplete search terms. As the basis for a systematic mapping is the definition of the search terms, we derived the terms of interest from the main search question and performed various cross-reviews and testing of terms among the authors.

Publication bias. As positive results are more likely to be published than negative ones, systematic literature mappings might suffer from this common bias. However, the findings of the articles, whether positive or negative, have solely a minor influence on this systematic mapping since we even included studies that did not report the results of their practical application.

Identification of studies. The main risk in a systematic literature mapping is the omission of relevant studies. In order to include as many studies as possible, we have chosen not to be strictly rigorous in defining automated process assessment, opening the possibility to include both partially automated studies and others that only indicate automation practices, even conceptual ones. As this research refers to the automation of process assessment, typically involving tool support, it is possible to note the high predominance of such experiences in the software-development industry. This behavior could already have been expected, most likely due to the influence of the words used in the search string, where “tool” and “automation” appear.

Selection and data extraction of studies. Threats to study selection and data extraction have been mitigated through a detailed definition of the inclusion/exclusion criteria and rigorous peer reviews among the authors.

6 CONCLUSIONS

This paper presents a systematic literature mapping aiming to identify the state of the art in automated process assessment. The literature mapping was performed through the definition of a research protocol, search execution, study selection, data extraction and synthesis. In the 16 studies found, that directly or indirectly proposes some kind of automation to support process assessment, several positive results were observed, with special emphasis on time and effort reduction and facilitating processes assessment. Some of the primary studies found, however, are considerably old (2002, for instance) to be considered as in the state of the art.

Although most studies do not cover all the typical activities expected from a process assessment, most propose the automation of at least part of them. As expected, software tools are used by most studies to support process assessment. However, the minority of studies only merely propose some approach for automated assessment. Several process reference models are used for automated processes assessment, with predominance of typical software-related models such as CMMI, Scrum and ISO standards. Also, among those who modelled the process to be assessed, most studies used a classical BPMN approach.

Future work indicates the need for greater coverage of tools and techniques to support process assessment. We plan to continue the evolution of the integration between BPM and Information Systems, contributing to the adoption and evolution of process-aware information systems.

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